

V3 parameter estimation

Six models · SPY, AAPL, MSFT · four windows · 1.5-year monthly rolling

What this report is

- The monthly rolling parameter paths used in the 1.5-year study, summarized by regime. One table per model.
- No LSM, no option prices, and no simulated paths. Only calibrate_ticker with the 18-month lookback.
- The companion notebook is the computational source of truth. This PDF is written from the same payload.

How estimates are formed

- At each month-end (plus the first trading day of the regime), parameters use only returns and listed quotes with date $\leq$ that update.
- Lookback = 18 months. Rolling = monthly. Same four windows as the decision study.
- Cells are Mean, SD, Min, and Max of those monthly estimates inside the regime.

Tables

- P1 GBM (μ, σ). P2 GARCH ($\mu, \omega, \alpha, \beta, \sigma_0$). P3 Heston ($\mu, \kappa, \theta, \xi, \rho, \nu_0$).
- P4 Merton ($\mu, \sigma, \lambda, \mu_J, \sigma_J, \kappa$). P5 GARCH-Merton. P6 Heston-Merton.
- R1 Crisis, R2 Normal, R3 Late-cycle, R4 COVID. Columns are SPY, AAPL, MSFT.

Table P1 · GBM · monthly rolling parameter estimates

1.5-year lookback · monthly updates · Mean, SD, Min, Max within each regime

Regime / parameter	SPY				AAPL				MSFT			
Regime / parameter	Mean	SD	Min	Max	Mean	SD	Min	Max	Mean	SD	Min	Max
R1 Crisis (n=18)												
μ	-0.2873	0.1250	-0.4424	-0.0608	-0.0499	0.2597	-0.2648	0.4609	-0.2249	0.1181	-0.3572	-0.0067
σ	0.3262	0.0777	0.1852	0.3895	0.5106	0.0533	0.4040	0.5490	0.4089	0.0803	0.2730	0.4812
R2 Normal (n=18)												
μ	0.1779	0.0221	0.1476	0.2233	0.1615	0.2230	-0.1422	0.4632	0.2651	0.0662	0.1597	0.3537
σ	0.1120	0.0031	0.1075	0.1165	0.2615	0.0317	0.2210	0.2953	0.2336	0.0039	0.2276	0.2404
R3 Late-cycle (n=18)												
μ	0.0701	0.0323	0.0222	0.1419	0.1466	0.0860	0.0279	0.3211	0.3024	0.0364	0.2586	0.3944
σ	0.1452	0.0153	0.1100	0.1582	0.2689	0.0286	0.2073	0.2922	0.2477	0.0186	0.2007	0.2629
R4 COVID (n=18)												
μ	0.0766	0.0617	-0.0804	0.1512	0.3347	0.2027	0.0935	0.7360	0.3438	0.0843	0.2293	0.4846
σ	0.2108	0.0695	0.1452	0.2866	0.3314	0.0460	0.2857	0.3935	0.2962	0.0596	0.2366	0.3640

Notes: This table reports Mean, SD, Min, and Max of the monthly rolling GBM estimates for SPY, AAPL, and MSFT. Parameters are μ , σ . Each month-end update uses an 18-month lookback ending on that date, with no look-ahead. R1-R4 are Crisis, Normal, Late-cycle, and COVID. n is the number of monthly updates in that regime. SD is the sample standard deviation across those monthly estimates. This is parameter estimation only; it is not an LSM pricing table.

Table P2 · GARCH · monthly rolling parameter estimates

1.5-year lookback · monthly updates · Mean, SD, Min, Max within each regime

Regime / parameter	SPY				AAPL				MSFT			
Regime / parameter	Mean	SD	Min	Max	Mean	SD	Min	Max	Mean	SD	Min	Max
R1 Crisis (n=18)												
μ	-0.1456	0.0959	-0.2885	0.0217	0.4332	0.2065	0.2063	0.7693	-0.1228	0.0870	-0.2561	0.0147
ω	5.787e-06	1.579e-06	2.695e-06	7.491e-06	4.737e-05	1.436e-05	1.423e-05	6.200e-05	2.186e-05	7.284e-06	1.204e-05	3.168e-05
α	0.1105	0.0210	0.0554	0.1310	0.1219	0.0165	0.0985	0.1526	0.1665	0.0465	0.1011	0.2265
β	0.8777	0.0191	0.8596	0.9275	0.8346	0.0255	0.8048	0.8787	0.8192	0.0363	0.7715	0.8829
σ_0	0.0252	0.0123	0.0125	0.0488	0.0344	0.0190	0.0174	0.0864	0.0326	0.0125	0.0151	0.0542
R2 Normal (n=18)												
μ	0.2085	0.0194	0.1720	0.2434	0.2102	0.1835	-0.0765	0.4569	0.4654	0.1294	0.2948	0.6676
ω	1.095e-05	2.978e-06	6.320e-06	1.489e-05	2.811e-06	4.526e-06	4.586e-10	1.103e-05	1.043e-04	3.348e-05	5.913e-05	1.419e-04
α	0.1503	0.0455	0.1052	0.2316	0.0089	0.0142	1.206e-14	0.0382	0.3332	0.1472	0.1734	0.5488
β	0.6339	0.0780	0.4930	0.7198	0.9827	0.0256	0.9434	0.9996	0.2917	0.0636	0.2263	0.3889
σ_0	0.0064	8.979e-04	0.0050	0.0083	0.0146	0.0024	0.0126	0.0211	0.0136	0.0015	0.0106	0.0171
R3 Late-cycle (n=18)												
μ	0.2628	0.0346	0.2145	0.3268	0.4107	0.1191	0.2528	0.5870	0.4559	0.0538	0.3583	0.5498
ω	3.229e-06	7.878e-07	2.130e-06	4.319e-06	2.252e-05	1.066e-05	1.098e-05	4.560e-05	2.951e-05	9.317e-06	1.488e-05	4.575e-05
α	0.2215	0.0171	0.2006	0.2440	0.1595	0.0263	0.1152	0.1931	0.2156	0.0326	0.1338	0.2662
β	0.7618	0.0206	0.7197	0.7898	0.7631	0.0661	0.6154	0.8558	0.6631	0.0758	0.5131	0.8008
σ_0	0.0094	0.0051	0.0044	0.0225	0.0168	0.0060	0.0108	0.0289	0.0154	0.0050	0.0096	0.0268
R4 COVID (n=18)												
μ	0.2990	0.0538	0.2105	0.3683	0.7551	0.1424	0.5243	0.9934	0.4506	0.0621	0.3583	0.5217
ω	4.354e-06	4.770e-07	3.195e-06	5.362e-06	1.507e-05	2.480e-06	1.119e-05	1.916e-05	1.150e-05	3.536e-06	6.712e-06	1.988e-05
α	0.2464	0.0598	0.1658	0.3105	0.1802	0.0330	0.1476	0.2469	0.1941	0.0494	0.1337	0.2689
β	0.7321	0.0400	0.6875	0.7903	0.7996	0.0225	0.7517	0.8315	0.7770	0.0300	0.7289	0.8098
σ_0	0.0142	0.0128	0.0052	0.0493	0.0214	0.0120	0.0118	0.0544	0.0204	0.0148	0.0091	0.0611

Notes: This table reports Mean, SD, Min, and Max of the monthly rolling GARCH estimates for SPY, AAPL, and MSFT. Parameters are μ , ω , α , β , σ_0 . Each month-end update uses an 18-month lookback ending on that date, with no look-ahead. R1-R4 are Crisis, Normal, Late-cycle, and COVID. n is the number of monthly updates in that regime. SD is the sample standard deviation across those monthly estimates. This is parameter estimation only; it is not an LSM pricing table.

Table P3 · Heston · monthly rolling parameter estimates

1.5-year lookback · monthly updates · Mean, SD, Min, Max within each regime

Regime / param	SPY				AAPL				MSFT			
Regime / param	Mean	SD	Min	Max	Mean	SD	Min	Max	Mean	SD	Min	Max
R1 Crisis (n =												
μ	-0.2873	0.1250	-0.4424	-0.0608	-0.0499	0.2597	-0.2648	0.4609	-0.2249	0.1181	-0.3572	-0.0067
κ	12.0625	6.0476	4.8057	20.0000	8.5388	7.2924	0.7342	20.0000	9.4955	7.6740	0.8851	20.0000
θ	0.1518	0.0928	0.0523	0.2891	0.6317	0.4968	0.1534	1.5768	0.3112	0.2556	1.001e-04	0.7698
ξ	2.3229	0.9505	0.6999	4.0000	2.4036	0.8608	1.4865	4.0000	2.1593	0.8496	0.5021	4.0000
ρ	-0.6503	0.0805	-0.8449	-0.5025	-0.4567	0.1253	-0.7573	-0.3228	-0.3745	0.1169	-0.6165	-0.2109
v ₀	0.1655	0.1320	0.0243	0.5054	0.2694	0.2001	0.0623	0.6809	0.1713	0.0871	0.0773	0.3917
R2 Normal (n												
μ	0.1779	0.0221	0.1476	0.2233	0.1615	0.2230	-0.1422	0.4632	0.2651	0.0662	0.1597	0.3537
κ	12.8272	8.1134	1.3119	20.0000	18.9598	2.8433	9.7128	20.0000	17.3446	6.1856	1.8845	20.0000
θ	0.0308	0.0188	0.0102	0.0770	0.0782	0.0343	0.0281	0.1505	0.0380	0.0181	1.000e-04	0.0694
ξ	1.6010	0.7922	0.6762	3.4642	2.7159	0.7042	1.5709	4.0000	1.3628	0.8289	0.0833	2.6476
ρ	-0.6464	0.1295	-0.8118	-0.3637	-0.2014	0.1434	-0.4077	0.0467	-0.2202	0.4252	-0.9900	0.9900
v ₀	0.0176	0.0088	0.0072	0.0380	0.0607	0.0645	0.0034	0.2669	0.0399	0.0166	0.0198	0.0791
R3 Late-cycle												
μ	0.0701	0.0323	0.0222	0.1419	0.1466	0.0860	0.0279	0.3211	0.3024	0.0364	0.2586	0.3944
κ	17.0936	5.6562	1.4735	20.0000	18.9355	2.4866	12.0825	20.0000	19.5573	1.0280	16.6535	20.0000
θ	0.0347	0.0198	0.0133	0.0879	0.0870	0.0571	0.0186	0.2082	0.0659	0.0367	1.000e-04	0.1415
ξ	2.2330	1.0300	0.9308	4.0000	3.6065	0.5957	2.2325	4.0000	2.5010	0.9028	1.3339	4.0000
ρ	-0.7028	0.0698	-0.8074	-0.6013	-0.4579	0.1114	-0.6065	-0.1969	-0.4721	0.1348	-0.6163	-0.1178
v ₀	0.0404	0.0315	0.0078	0.1043	0.1618	0.1009	0.0366	0.3439	0.0917	0.0742	0.0176	0.2375
R4 COVID (n :												
μ	0.0766	0.0617	-0.0804	0.1512	0.3347	0.2027	0.0935	0.7360	0.3438	0.0843	0.2293	0.4846
κ	14.4265	6.3738	1.0387	20.0000	18.0413	4.1722	5.6711	20.0000	18.8111	2.6664	10.8491	20.0000
θ	0.1177	0.2003	0.0134	0.7646	0.1102	0.0563	0.0106	0.2370	0.0992	0.0518	0.0213	0.1914
ξ	2.6785	0.9682	1.6644	4.0000	3.6105	0.8181	1.5009	4.0000	3.2837	0.7114	2.1818	4.0000
ρ	-0.7455	0.1034	-0.9900	-0.5732	-0.4122	0.2175	-0.6941	0.1872	-0.4580	0.1520	-0.6506	-0.1726
v ₀	0.0961	0.1787	0.0076	0.6776	0.2057	0.1584	0.0550	0.5548	0.1660	0.1915	0.0272	0.7179

Notes: This table reports Mean, SD, Min, and Max of the monthly rolling Heston estimates for SPY, AAPL, and MSFT. Parameters are μ, κ, θ, ξ, ρ, v₀. Each month-end update uses an 18-month lookback ending on that date, with no look-ahead. R1-R4 are Crisis, Normal, Late-cycle, and COVID. n is the number of monthly updates in that regime. SD is the sample standard deviation across those monthly estimates. This is parameter estimation only; it is not an LSM pricing table.

Table P4 · Merton · monthly rolling parameter estimates

1.5-year lookback · monthly updates · Mean, SD, Min, Max within each regime

Regime / param	SPY				AAPL				MSFT			
Regime / param	Mean	SD	Min	Max	Mean	SD	Min	Max	Mean	SD	Min	Max
R1 Crisis (n =												
μ	-0.2184	0.0900	-0.3501	-0.0707	0.0013	0.2783	-0.2261	0.5259	-0.3118	0.2108	-0.5198	0.0731
σ	0.2709	0.0606	0.1725	0.3288	0.4442	0.0414	0.3678	0.4777	0.3424	0.0678	0.2379	0.4109
λ	3.8907	1.0064	1.3263	5.3053	2.5088	0.2923	1.9895	2.6880	3.5830	0.5718	2.6526	4.6421
μ_J	-0.0146	0.0205	-0.0308	0.0388	-0.0202	0.0200	-0.0750	0.0170	0.0211	0.0269	-0.0259	0.0485
σ_J	0.0833	0.0307	0.0036	0.1011	0.1554	0.0222	0.1145	0.1858	0.1130	0.0218	0.0778	0.1328
κ	-0.0105	0.0191	-0.0261	0.0396	-0.0076	0.0221	-0.0649	0.0349	0.0284	0.0293	-0.0226	0.0559
R2 Normal (n												
μ	0.2198	0.0276	0.1728	0.2727	0.2594	0.1258	0.0754	0.4440	0.3161	0.0715	0.2188	0.4153
σ	0.1016	0.0037	0.0965	0.1065	0.2081	0.0277	0.1714	0.2411	0.1796	0.0048	0.1723	0.1884
λ	3.5304	0.6309	1.9791	4.6667	4.2451	0.5021	3.3245	4.6790	3.1212	0.3254	2.6387	3.3511
μ_J	-0.0108	0.0043	-0.0153	-0.0049	-0.0245	0.0288	-0.0541	0.0059	-0.0147	0.0034	-0.0169	-0.0098
σ_J	0.0223	0.0031	0.0182	0.0264	0.0687	0.0050	0.0644	0.0791	0.0835	0.0051	0.0802	0.0908
κ	-0.0105	0.0043	-0.0150	-0.0045	-0.0215	0.0280	-0.0505	0.0080	-0.0112	0.0038	-0.0136	-0.0057
R3 Late-cycle												
μ	0.1458	0.0444	0.0916	0.2578	0.1514	0.0680	0.0507	0.2592	0.2827	0.0605	0.2005	0.4313
σ	0.1201	0.0139	0.0866	0.1321	0.2268	0.0225	0.1723	0.2454	0.2090	0.0204	0.1625	0.2270
λ	4.6249	0.3420	3.9685	5.3333	4.0056	0.7597	3.3333	6.0000	4.2127	0.6868	2.6667	5.3475
μ_J	-0.0159	0.0049	-0.0239	-0.0107	-0.0011	0.0081	-0.0096	0.0164	0.0058	0.0073	-0.0075	0.0170
σ_J	0.0337	0.0060	0.0208	0.0377	0.0727	0.0140	0.0440	0.0836	0.0642	0.0024	0.0585	0.0676
κ	-0.0152	0.0049	-0.0233	-0.0100	0.0017	0.0079	-0.0079	0.0175	0.0079	0.0074	-0.0058	0.0194
R4 COVID (n :												
μ	0.1067	0.0647	-0.0407	0.1676	0.3576	0.1562	0.1487	0.6455	0.3393	0.0852	0.2152	0.4715
σ	0.1516	0.0309	0.1210	0.1903	0.2613	0.0222	0.2376	0.3003	0.2321	0.0223	0.2072	0.2662
λ	5.2356	0.6039	4.6421	6.0319	4.6200	0.7491	3.9789	6.0000	4.3121	1.2686	2.6667	6.0000
μ_J	-0.0057	0.0078	-0.0172	0.0075	-0.0047	0.0137	-0.0278	0.0252	0.0033	0.0097	-0.0148	0.0170
σ_J	0.0607	0.0285	0.0335	0.0920	0.0922	0.0156	0.0751	0.1118	0.0838	0.0225	0.0615	0.1122
κ	-0.0034	0.0094	-0.0165	0.0118	-2.834e-04	0.0143	-0.0247	0.0308	0.0071	0.0087	-0.0099	0.0194

Notes: This table reports Mean, SD, Min, and Max of the monthly rolling Merton estimates for SPY, AAPL, and MSFT. Parameters are μ , σ , λ , μ_J , σ_J , κ . Each month-end update uses an 18-month lookback ending on that date, with no look-ahead. R1-R4 are Crisis, Normal, Late-cycle, and COVID. n is the number of monthly updates in that regime. SD is the sample standard deviation across those monthly estimates. This is parameter estimation only; it is not an LSM pricing table.

Table P5 · GARCH-Merton · monthly rolling parameter estimates

1.5-year lookback · monthly updates · Mean, SD, Min, Max within each regime

Regime / param	SPY				AAPL				MSFT			
Regime / param	Mean	SD	Min	Max	Mean	SD	Min	Max	Mean	SD	Min	Max
R1 Crisis (n =												
μ	-0.1456	0.0959	-0.2885	0.0217	0.4332	0.2065	0.2063	0.7693	-0.1228	0.0870	-0.2561	0.0147
ω	5.787e-06	1.579e-06	2.695e-06	7.491e-06	4.737e-05	1.436e-05	1.423e-05	6.200e-05	2.186e-05	7.284e-06	1.204e-05	3.168e-05
α	0.1105	0.0210	0.0554	0.1310	0.1219	0.0165	0.0985	0.1526	0.1665	0.0465	0.1011	0.2265
β	0.8777	0.0191	0.8596	0.9275	0.8346	0.0255	0.8048	0.8787	0.8192	0.0363	0.7715	0.8829
σ_0	0.0252	0.0123	0.0125	0.0488	0.0344	0.0190	0.0174	0.0864	0.0326	0.0125	0.0151	0.0542
λ	0.7678	0.2487	0.6597	1.3298	2.1505	0.4001	1.3333	2.6880	3.1250	0.7482	2.0000	4.0320
μ_J	-0.0801	0.0233	-0.0919	-0.0285	-0.1202	0.0177	-0.1438	-0.0851	0.0121	0.0135	-0.0138	0.0255
σ_J	0.0039	0.0129	0.0000	0.0467	0.0549	0.0143	0.0251	0.0759	0.1149	0.0175	0.0778	0.1262
κ	-0.0767	0.0220	-0.0878	-0.0281	-0.1117	0.0153	-0.1314	-0.0813	0.0192	0.0154	-0.0096	0.0340
R2 Normal (n =												
μ	0.2085	0.0194	0.1720	0.2434	0.2102	0.1835	-0.0765	0.4569	0.4654	0.1294	0.2948	0.6676
ω	1.095e-05	2.978e-06	6.320e-06	1.489e-05	2.811e-06	4.526e-06	4.586e-10	1.103e-05	1.043e-04	3.348e-05	5.913e-05	1.419e-04
α	0.1503	0.0455	0.1052	0.2316	0.0089	0.0142	1.206e-14	0.0382	0.3332	0.1472	0.1734	0.5488
β	0.6339	0.0780	0.4930	0.7198	0.9827	0.0256	0.9434	0.9996	0.2917	0.0636	0.2263	0.3889
σ_0	0.0064	8.979e-04	0.0050	0.0083	0.0146	0.0024	0.0126	0.0211	0.0136	0.0015	0.0106	0.0171
λ	2.3531	0.7474	1.3194	3.3245	4.9102	0.4997	3.9894	5.9528	2.8644	0.3150	2.6387	3.3333
μ_J	-0.0206	0.0033	-0.0231	-0.0128	-0.0177	0.0184	-0.0392	0.0040	-0.0069	0.0046	-0.0098	-3.403e-04
σ_J	0.0045	0.0073	6.892e-04	0.0220	0.0699	0.0071	0.0608	0.0801	0.0879	0.0045	0.0814	0.0908
κ	-0.0203	0.0033	-0.0228	-0.0124	-0.0150	0.0177	-0.0359	0.0062	-0.0030	0.0042	-0.0057	0.0030
R3 Late-cycle (n =												
μ	0.2628	0.0346	0.2145	0.3268	0.4107	0.1191	0.2528	0.5870	0.4559	0.0538	0.3583	0.5498
ω	3.229e-06	7.878e-07	2.130e-06	4.319e-06	2.252e-05	1.066e-05	1.098e-05	4.560e-05	2.951e-05	9.317e-06	1.488e-05	4.575e-05
α	0.2215	0.0171	0.2006	0.2440	0.1595	0.0263	0.1152	0.1931	0.2156	0.0326	0.1338	0.2662
β	0.7618	0.0206	0.7197	0.7898	0.7631	0.0661	0.6154	0.8558	0.6631	0.0758	0.5131	0.8008
σ_0	0.0094	0.0051	0.0044	0.0225	0.0168	0.0060	0.0108	0.0289	0.0154	0.0050	0.0096	0.0268
λ	3.1839	0.6655	2.0000	4.6299	4.3171	0.8081	3.3071	5.3760	3.0309	0.3434	2.6596	3.3600
μ_J	-0.0289	0.0021	-0.0312	-0.0241	-0.0174	0.0069	-0.0237	-0.0029	-0.0103	0.0143	-0.0431	0.0169
σ_J	0.0083	0.0020	0.0042	0.0107	0.0638	0.0074	0.0502	0.0705	0.0585	0.0137	0.0143	0.0647
κ	-0.0285	0.0020	-0.0307	-0.0238	-0.0152	0.0064	-0.0210	-0.0017	-0.0083	0.0145	-0.0421	0.0188
R4 COVID (n =												
μ	0.2990	0.0538	0.2105	0.3683	0.7551	0.1424	0.5243	0.9934	0.4506	0.0621	0.3583	0.5217
ω	4.354e-06	4.770e-07	3.195e-06	5.362e-06	1.507e-05	2.480e-06	1.119e-05	1.916e-05	1.150e-05	3.536e-06	6.712e-06	1.988e-05
α	0.2464	0.0598	0.1658	0.3105	0.1802	0.0330	0.1476	0.2469	0.1941	0.0494	0.1337	0.2689
β	0.7321	0.0400	0.6875	0.7903	0.7996	0.0225	0.7517	0.8315	0.7770	0.0300	0.7289	0.8098
σ_0	0.0142	0.0128	0.0052	0.0493	0.0214	0.0120	0.0118	0.0544	0.0204	0.0148	0.0091	0.0611
λ	2.9759	0.4353	2.0214	3.3511	4.1561	1.2477	2.0000	5.3333	3.1818	0.6170	2.6526	4.0213
μ_J	-0.0318	0.0031	-0.0372	-0.0292	-0.0242	0.0070	-0.0329	-0.0136	-0.0282	0.0091	-0.0431	-0.0095
σ_J	0.0067	0.0053	0.0036	0.0160	0.0700	0.0068	0.0635	0.0886	0.0377	0.0132	0.0143	0.0613
κ	-0.0313	0.0029	-0.0364	-0.0288	-0.0215	0.0068	-0.0291	-0.0106	-0.0270	0.0093	-0.0421	-0.0081

Notes: This table reports Mean, SD, Min, and Max of the monthly rolling GARCH-Merton estimates for SPY, AAPL, and MSFT. Parameters are μ , ω , α , β , σ_0 , λ , μ_J , σ_J , κ . Each month-end update uses an 18-month lookback ending on that date, with no look-ahead. R1-R4 are Crisis, Normal, Late-cycle, and COVID. n is the number of monthly updates in that regime. SD is the sample standard deviation across those monthly estimates. This is parameter estimation only; it is not an LSM pricing table.

Table P6 · Heston-Merton · monthly rolling parameter estimates

1.5-year lookback · monthly updates · Mean, SD, Min, Max within each regime

Regime / param	SPY				AAPL				MSFT			
Regime / param	Mean	SD	Min	Max	Mean	SD	Min	Max	Mean	SD	Min	Max
R1 Crisis (n =												
μ	-0.2873	0.1250	-0.4424	-0.0608	-0.0499	0.2597	-0.2648	0.4609	-0.2249	0.1181	-0.3572	-0.0067
κ	7.6185	6.2638	0.9153	20.0000	9.1102	7.4299	0.4376	20.0000	6.2663	5.0897	0.8393	20.0000
θ	0.1350	0.1251	0.0091	0.3838	0.5133	0.4908	0.0989	1.4416	0.2589	0.3275	1.000e-04	1.0396
ξ	1.5434	1.0236	0.4478	4.0000	1.7096	1.1643	0.4852	4.0000	1.5312	0.9853	0.1393	4.0000
ρ	-0.8580	0.1188	-0.9900	-0.6881	-0.7930	0.2106	-0.9900	-0.4626	-0.8558	0.2049	-0.9900	-0.3892
ν_0	0.1287	0.1221	0.0165	0.4470	0.2273	0.1915	0.0309	0.6331	0.1331	0.0834	0.0514	0.3456
λ	3.8907	1.0064	1.3263	5.3053	2.5088	0.2923	1.9895	2.6880	3.5830	0.5718	2.6526	4.6421
μ_J	-0.0146	0.0205	-0.0308	0.0388	-0.0202	0.0200	-0.0750	0.0170	0.0211	0.0269	-0.0259	0.0485
σ_J	0.0833	0.0307	0.0036	0.1011	0.1554	0.0222	0.1145	0.1858	0.1130	0.0218	0.0778	0.1328
κ_J	-0.0105	0.0191	-0.0261	0.0396	-0.0076	0.0221	-0.0649	0.0349	0.0284	0.0293	-0.0226	0.0559
R2 Normal (n =												
μ	0.1779	0.0221	0.1476	0.2233	0.1615	0.2230	-0.1422	0.4632	0.2651	0.0662	0.1597	0.3537
κ	11.2003	8.2582	1.2924	20.0000	19.0080	2.4977	11.2688	20.0000	15.6890	7.7563	0.1405	20.0000
θ	0.0304	0.0189	0.0065	0.0663	0.0440	0.0347	1.000e-04	0.1130	0.0272	0.0590	1.000e-04	0.2193
ξ	1.8878	1.0999	0.6272	4.0000	1.3449	0.9821	0.2892	4.0000	0.4144	0.3774	0.0627	1.4945
ρ	-0.6154	0.2486	-0.8284	-0.0609	-0.1212	0.7127	-0.9900	0.9900	-0.6998	0.6148	-0.9900	0.9900
ν_0	0.0167	0.0082	0.0065	0.0353	0.0423	0.0593	1.000e-04	0.2342	0.0284	0.0141	0.0117	0.0627
λ	3.5304	0.6309	1.9791	4.6667	4.2451	0.5021	3.3245	4.6790	3.1212	0.3254	2.6387	3.3511
μ_J	-0.0108	0.0043	-0.0153	-0.0049	-0.0245	0.0288	-0.0541	0.0059	-0.0147	0.0034	-0.0169	-0.0098
σ_J	0.0223	0.0031	0.0182	0.0264	0.0687	0.0050	0.0644	0.0791	0.0835	0.0051	0.0802	0.0908
κ_J	-0.0105	0.0043	-0.0150	-0.0045	-0.0215	0.0280	-0.0505	0.0080	-0.0112	0.0038	-0.0136	-0.0057
R3 Late-cycle												
μ	0.0701	0.0323	0.0222	0.1419	0.1466	0.0860	0.0279	0.3211	0.3024	0.0364	0.2586	0.3944
κ	13.2397	9.0334	0.1758	20.0000	19.3774	1.6402	14.3824	20.0000	16.8793	5.2952	5.5175	20.0000
θ	0.1140	0.2083	0.0020	0.7622	0.0560	0.0527	1.000e-04	0.1898	0.0453	0.0395	1.000e-04	0.1164
ξ	2.9064	1.2657	1.0544	4.0000	2.6151	1.3779	0.9235	4.0000	1.7638	1.1302	0.6691	4.0000
ρ	-0.7722	0.1922	-0.9486	-0.2763	-0.7259	0.2419	-0.9900	-0.3627	-0.8317	0.2700	-0.9900	-0.1707
ν_0	0.0388	0.0302	0.0046	0.1009	0.1414	0.1087	0.0040	0.3330	0.0739	0.0718	0.0057	0.2127
λ	4.6249	0.3420	3.9685	5.3333	4.0056	0.7597	3.3333	6.0000	4.2127	0.6868	2.6667	5.3475
μ_J	-0.0159	0.0049	-0.0239	-0.0107	-0.0011	0.0081	-0.0096	0.0164	0.0058	0.0073	-0.0075	0.0170
σ_J	0.0337	0.0060	0.0208	0.0377	0.0727	0.0140	0.0440	0.0836	0.0642	0.0024	0.0585	0.0676
κ_J	-0.0152	0.0049	-0.0233	-0.0100	0.0017	0.0079	-0.0079	0.0175	0.0079	0.0074	-0.0058	0.0194
R4 COVID (n =												
μ	0.0766	0.0617	-0.0804	0.1512	0.3347	0.2027	0.0935	0.7360	0.3438	0.0843	0.2293	0.4846
κ	12.7000	9.1903	0.1931	20.0000	19.2715	1.7657	13.8249	20.0000	12.2117	8.6272	0.0500	20.0000
θ	0.1062	0.1630	1.000e-04	0.5512	0.0548	0.0427	1.000e-04	0.1535	0.1704	0.2305	1.000e-04	0.7721
ξ	2.5865	1.2838	0.2116	4.0000	2.7938	1.0559	1.4037	4.0000	1.9181	1.4008	0.4946	4.0000
ρ	-0.9421	0.0820	-0.9900	-0.6842	-0.7080	0.3386	-0.9900	0.0996	-0.8580	0.2188	-0.9900	-0.3549
ν_0	0.0857	0.1693	0.0035	0.6385	0.1736	0.1536	0.0326	0.5103	0.1346	0.1885	0.0141	0.6761
λ	5.2356	0.6039	4.6421	6.0319	4.6200	0.7491	3.9789	6.0000	4.3121	1.2686	2.6667	6.0000
μ_J	-0.0057	0.0078	-0.0172	0.0075	-0.0047	0.0137	-0.0278	0.0252	0.0033	0.0097	-0.0148	0.0170
σ_J	0.0607	0.0285	0.0335	0.0920	0.0922	0.0156	0.0751	0.1118	0.0838	0.0225	0.0615	0.1122
κ_J	-0.0034	0.0094	-0.0165	0.0118	-2.834e-04	0.0143	-0.0247	0.0308	0.0071	0.0087	-0.0099	0.0194

Notes: This table reports Mean, SD, Min, and Max of the monthly rolling Heston-Merton estimates for SPY, AAPL, and MSFT. Parameters are μ , κ , θ , ξ , ρ , ν_0 , λ , μ_J , σ_J , κ_J . Each month-end update uses an 18-month lookback ending on that date, with no look-ahead. R1-R4 are Crisis, Normal, Late-cycle, and COVID. n is the number of monthly updates in that regime. SD is the sample standard deviation across those monthly estimates. This is parameter estimation only; it is not an LSM pricing table.